

Hamid Khan

Full-Stack Developer & AI Engineer — ERP & Business Systems

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SUMMARY

Full-stack developer and AI engineer specialising in **business and ERP systems** and **practical AI integration**. I build complete products end to end — database schema, REST API, authentication and permissions, frontend, and desktop packaging — with a consistent rule: **no fake functionality**. If a feature exists, it is wired to a real implementation and real data. Recent work spans four production-architected systems: an AI voice receptionist that books real appointments, a Windows desktop AI assistant with a governed tool-permission engine, a multi-role law firm ERP, and a university ERP with student, faculty, and admin portals.

TECHNICAL SKILLS

Backend	Node.js, Express, Python, FastAPI, REST API design, WebSockets
Frontend	JavaScript (ES6+), HTML5, CSS3, responsive UI
Databases	PostgreSQL, SQLite — schema design, migrations, seeded fixtures, parameterised query hygiene
AI Engineering	Anthropic SDK, tool-calling / agent loops, OpenAI-compatible providers, system prompt and tool-schema design, graceful degradation without keys
Auth & Security	JWT, bcrypt, role-based access control, zod validation, rate limiting, helmet / CSP, SQL-injection prevention, row-level ownership enforced in SQL
Desktop	Electron packaging (Windows / macOS / Linux)
Tooling	Git & GitHub, npm, pip, Jest, environment and configuration management

PROJECTS

AI Dental Receptionist

Node.js • Express • SQLite / PostgreSQL • Anthropic tool-calling • JWT • Web Speech API • Jest

- Built a patient-facing voice and chat receptionist that books, cancels, and reschedules appointments against a **real database** — no invented availability or mock responses.
- Implemented an availability engine with **slot locking and double-booking prevention**, shared by the REST API, the voice widget, and the LLM tool layer through a single code path.
- Wired real Anthropic tool-calling so the AI calls the same service functions as direct API clients; designed graceful degradation to a clear "AI unavailable" state when no key is configured.
- Layered JWT authentication, rate limiting, a `NotificationService` abstraction (email / SMS / WhatsApp), and a Jest test suite over the business logic.

AI voice / chat booking system

J.A.R.V.I.S.

Personal AI operating system — Windows desktop

Python 3.11+ • FastAPI • WebSocket • Web Speech API • Anthropic / OpenAI-compatible providers

- Designed a modular desktop assistant that performs **real computer operations** — launching apps, managing files, monitoring system health, running gated shell commands — exposed through **36 tools** across app, file, system, web, memory, and developer categories.
- Built a **permission and risk engine** classifying every action by permission level (safe / confirm / always-confirm) and risk (low → critical), so irreversible actions can never be silently auto-run.
- Implemented a genuine offline **rule-based intent router** so the assistant is fully useful with no API key and no internet, with cloud providers swappable at runtime and nothing hard-coded.
- Added multilingual routing (English, Urdu, Roman Urdu), a live system HUD (CPU / RAM / disk / network / battery), voice with wake word and barge-in, and a drop-in skill system — with every tool call and approval written to an audit log.

LexOffice

Law firm management ERP

Node.js • Express • SQLite • JWT • bcrypt • Electron

- Re-architected a single-file, browser-IndexedDB demo into a **multi-user client-server system** with a real server-side database and server-verified login.
- Built a REST API and data layer covering clients, cases, and firm modules, with **role-based permissions** for Admin, Lawyer, Assistant, and Accountant.
- Implemented first-run seeding so a new install self-populates demo data and is usable immediately.
- Packaged the full stack as a cross-platform **Electron desktop application** (.exe / .dmg / .ApplImage).

University ERP System

Student • Faculty • Admin portals

Node.js • Express • PostgreSQL • JWT • bcrypt • zod • helmet

- Delivered three role-based portals plus a public admissions website, all reading and writing **live PostgreSQL data** — no sample data in the running system.
- Implemented attendance marking with duplicate-save prevention, results with automatic letter-grade computation, CGPA, transcripts (print / PDF), exam slips, invoices, hostel, and multi-step request workflows (retest, course drop, defer term, final degree).
- Secured the system end to end: bcrypt hashing, fully parameterised queries, zod validation on every input, rate-limited login and write endpoints, and a Content-Security-Policy via helmet.
- Enforced **row-level ownership in SQL** rather than the UI — a student can only read their own records, and faculty can only mark or grade students enrolled on their own courses.

EXPERIENCE

Independent Software Developer

2024 – Present

Self-directed projects — Lahore, Pakistan

- Designed and delivered **four full-stack systems end to end** — database schema, REST API, authentication and permissions, frontend, and desktop packaging — across ERP, AI assistant, and AI voice-booking domains.
- Built AI features with the **Anthropic SDK using real tool-calling agent loops**, wired to the same production service functions the REST API uses, including graceful degradation when no API key is configured.
- Applied the same security baseline on every project: bcrypt hashing, JWT auth, role-based access control, zod validation, rate limiting, and row-level ownership enforced in SQL rather than the UI.

[Job Title — if you have prior employment]

[Month YYYY – Month YYYY]

[Company Name] — [City]

- *What you built or improved, and the measurable result.*
- *What you owned and who you worked with.*

EDUCATION

BS Computer Science

2025 – 2029 (expected)

Superior University, Lahore

Currently in 2nd semester. Coursework in data structures, OOP, databases, and web development — applied in four self-directed full-stack projects below.

LANGUAGES

- **Urdu** — Native
 - **English** — Professional working proficiency
 - **Punjabi** — Conversational
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All four projects are open-source on GitHub — a live portfolio with full write-ups is linked above.